

24MC3035 - Prashant Ranjan

// **Solution 1**

```
#include <iostream>
```

```
using namespace std;
```

```
const int MAX_SIZE = 50;
```

```
class List {
```

```
private:
```

```
    int A[MAX_SIZE];
```

```
    int size;
```

```
public:
```

```
    List() : size(0) {}
```

```
    void print() {
```

```
        cout << "[";
```

```
        for (int i = 0; i < size; i++) {
```

```
            cout << A[i];
```

```
            if (i < size - 1) cout << ", ";
```

```
        }
```

```
        cout << "]" << endl;
```

```
    }
```

```
    void insertAtEnd(int value) {
```

```
        if (size < MAX_SIZE) {
```

```
            A[size++] = value;
```

```
        }
```

```
    }
```

```
};
```

```
int main() {
```

```
    List list;
```

```
    cout << "Before 1: ";
```

```
    list.print();
```

```
    for (int i = 1; i <= 10; i++) {
```

```
        list.insertAtEnd(i * 10);
```

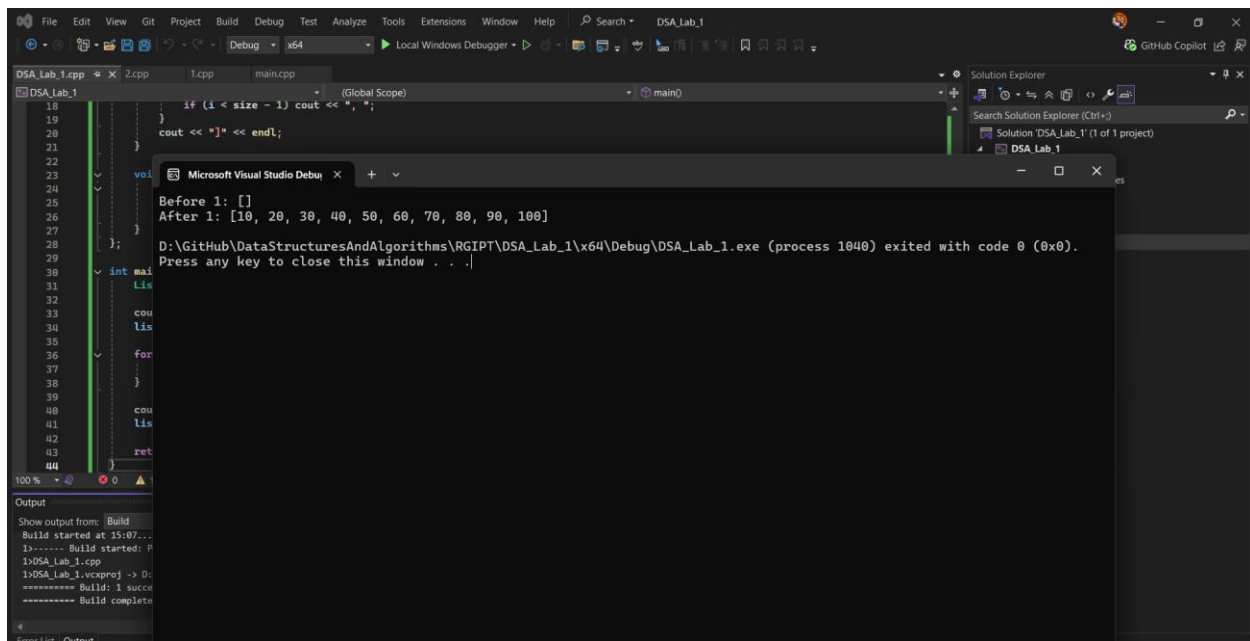
```
    }
```

```
    cout << "After 1: ";
```

```
    list.print();
```

```
    return 0;
```

```
}
```



// Solution 2

```
#include <iostream>
using namespace std;

const int MAX_SIZE = 50;

class List {
private:
    int A[MAX_SIZE];
    int size;

public:
    List() : size(0) {}

    void print() {
        cout << "[";
        for (int i = 0; i < size; i++) {
            cout << A[i];
            if (i < size - 1) cout << ", ";
        }
        cout << "]" << endl;
    }

    void insertAtEnd(int value) {
        if (size < MAX_SIZE) {
            A[size++] = value;
        }
    }

    void insertAtStart(int value) {
        if (size < MAX_SIZE) {
            for (int i = size; i > 0; i--) {
```

```

        A[i] = A[i - 1];
    }
    A[0] = value;
    size++;
}
}
};

int main() {
    List list;

    for (int i = 1; i <= 10; i++) {
        list.insertAtEnd(i * 10);
    }

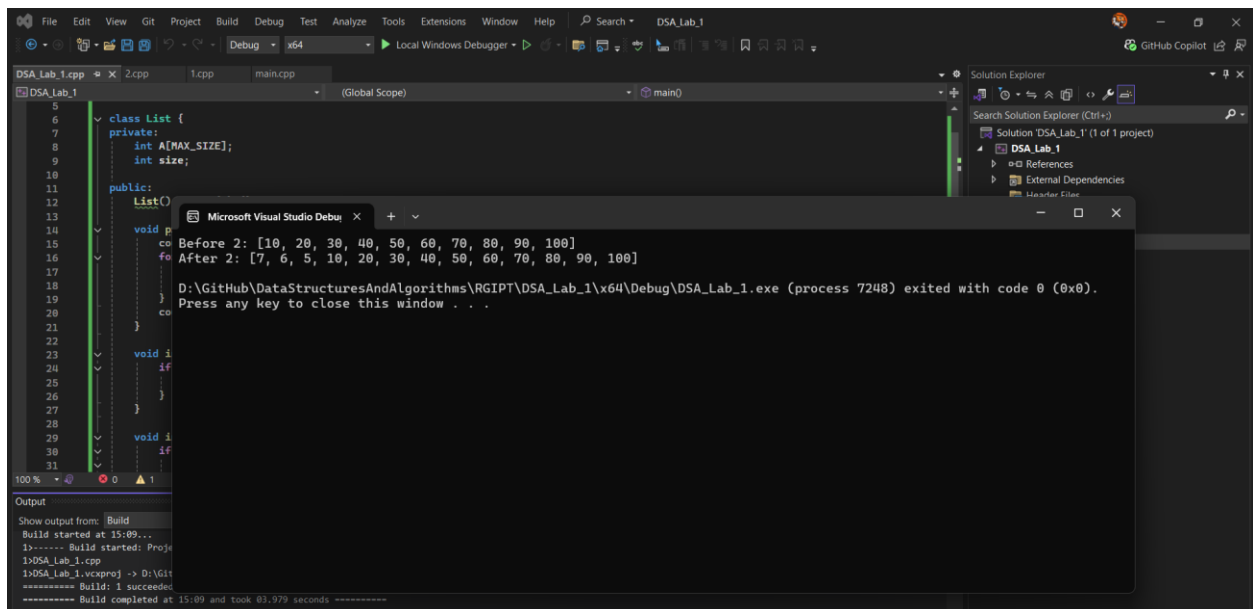
    cout << "Before 2: ";
    list.print();

    list.insertAtStart(5);
    list.insertAtStart(6);
    list.insertAtStart(7);

    cout << "After 2: ";
    list.print();

    return 0;
}

```



```

// Solution 3
#include <iostream>
using namespace std;

```

```

const int MAX_SIZE = 50;

class List {
private:
    int A[MAX_SIZE];
    int size;

public:
    List() : size(0) {}

    void print() {
        cout << "[";
        for (int i = 0; i < size; i++) {
            cout << A[i];
            if (i < size - 1) cout << ", ";
        }
        cout << "]" << endl;
    }

    void insertAtEnd(int value) {
        if (size < MAX_SIZE) {
            A[size++] = value;
        }
    }

    void insertAtStart(int value) {
        if (size < MAX_SIZE) {
            for (int i = size; i > 0; i--) {
                A[i] = A[i - 1];
            }
            A[0] = value;
            size++;
        }
    }

    void insertBefore(int beforeVal, int insertVal) {
        if (size >= MAX_SIZE) return;

        for (int i = 0; i < size; i++) {
            if (A[i] == beforeVal) {
                for (int j = size; j > i; j--) {
                    A[j] = A[j - 1];
                }
                A[i] = insertVal;
                size++;
                break;
            }
        }
    }
};

int main() {
    List list;

```

```

        for (int i = 1; i <= 10; i++) {
            list.insertAtEnd(i * 10);
        }
        list.insertAtStart(5);
        list.insertAtStart(6);
        list.insertAtStart(7);

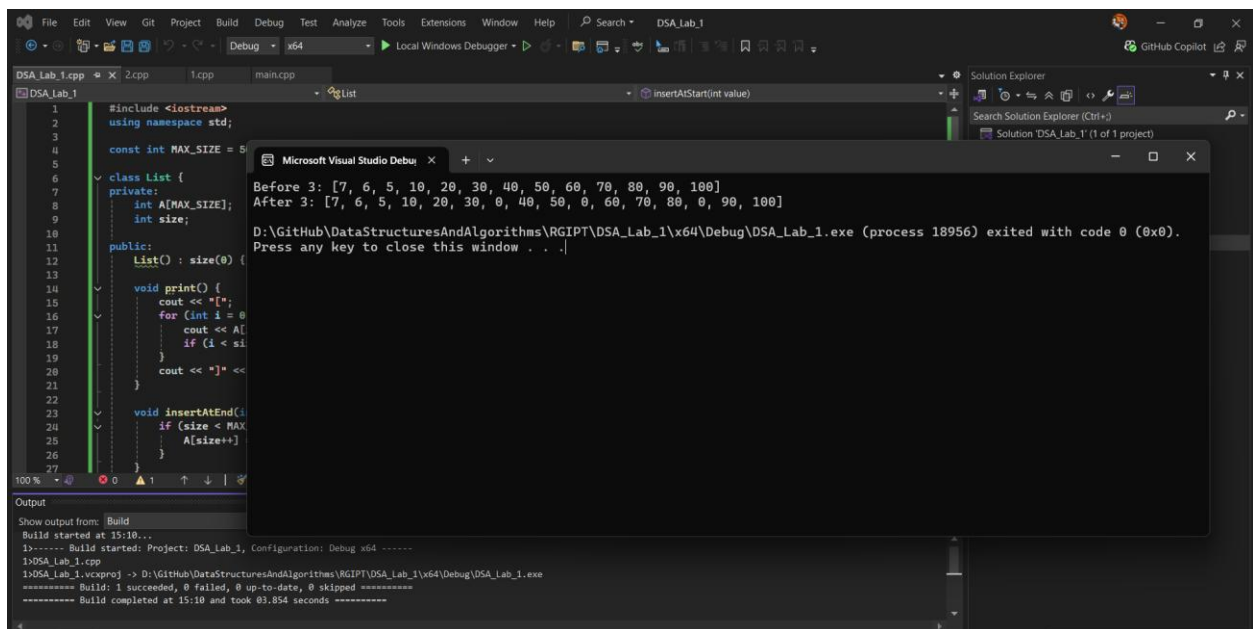
        cout << "Before 3: ";
        list.print();

        list.insertBefore(40, 0);
        list.insertBefore(60, 0);
        list.insertBefore(90, 0);

        cout << "After 3: ";
        list.print();

        return 0;
}

```



// Solution 4

```

#include <iostream>
using namespace std;

const int MAX_SIZE = 50;

class List {
private:
    int A[MAX_SIZE];

```

```

    int size;

public:
    List() : size(0) {}

    void print() {
        cout << "[";
        for (int i = 0; i < size; i++) {
            cout << A[i];
            if (i < size - 1) cout << ", ";
        }
        cout << "]" << endl;
    }

    void insertAtEnd(int value) {
        if (size < MAX_SIZE) {
            A[size++] = value;
        }
    }

    void insertAtStart(int value) {
        if (size < MAX_SIZE) {
            for (int i = size; i > 0; i--) {
                A[i] = A[i - 1];
            }
            A[0] = value;
            size++;
        }
    }

    void insertBefore(int beforeVal, int insertVal) {
        if (size >= MAX_SIZE) return;

        for (int i = 0; i < size; i++) {
            if (A[i] == beforeVal) {
                for (int j = size; j > i; j--) {
                    A[j] = A[j - 1];
                }
                A[i] = insertVal;
                size++;
                break;
            }
        }
    }

    void deleteAtStart() {
        if (size > 0) {
            for (int i = 0; i < size - 1; i++) {
                A[i] = A[i + 1];
            }
            size--;
        }
    }

```

```

        void deleteAtEnd() {
            if (size > 0) {
                size--;
            }
        }
    };

int main() {
    List list;
    for (int i = 1; i <= 10; i++) {
        list.insertAtEnd(i * 10);
    }
    list.insertAtStart(5);
    list.insertAtStart(6);
    list.insertAtStart(7);
    list.insertBefore(40, 0);
    list.insertBefore(60, 0);
    list.insertBefore(90, 0);

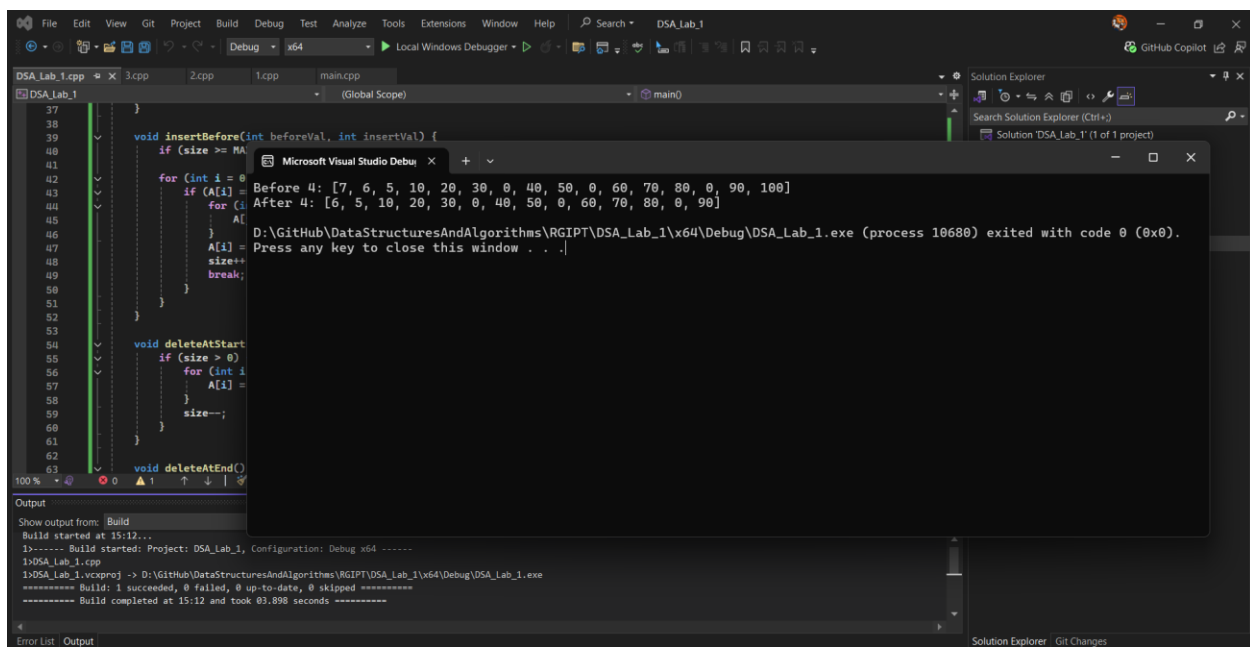
    cout << "Before 4: ";
    list.print();

    list.deleteAtStart();
    list.deleteAtEnd();

    cout << "After 4: ";
    list.print();

    return 0;
}

```



```

// Solution 5
#include <iostream>
using namespace std;

const int MAX_SIZE = 50;

class List {
private:
    int A[MAX_SIZE];
    int size;

public:
    List() : size(0) {}

    void print() {
        cout << "[";
        for (int i = 0; i < size; i++) {
            cout << A[i];
            if (i < size - 1) cout << ", ";
        }
        cout << "]" << endl;
    }

    void insertAtEnd(int value) {
        if (size < MAX_SIZE) {
            A[size++] = value;
        }
    }

    void insertAtStart(int value) {
        if (size < MAX_SIZE) {
            for (int i = size; i > 0; i--) {
                A[i] = A[i - 1];
            }
            A[0] = value;
            size++;
        }
    }

    void insertBefore(int beforeVal, int insertVal) {
        if (size >= MAX_SIZE) return;

        for (int i = 0; i < size; i++) {
            if (A[i] == beforeVal) {
                for (int j = size; j > i; j--) {
                    A[j] = A[j - 1];
                }
                A[i] = insertVal;
                size++;
                break;
            }
        }
    }
}

```



```

void deleteAtStart() {
    if (size > 0) {
        for (int i = 0; i < size - 1; i++) {
            A[i] = A[i + 1];
        }
        size--;
    }
}

void deleteAtEnd() {
    if (size > 0) {
        size--;
    }
}

void deleteBeforeAndAfter(int value) {
    for (int i = 0; i < size; i++) {
        if (A[i] == value) {
            // Delete after
            if (i < size - 1) {
                for (int j = i + 1; j < size - 1; j++) {
                    A[j] = A[j + 1];
                }
                size--;
            }

            // Delete before
            if (i > 0) {
                for (int j = i - 1; j < size - 1; j++) {
                    A[j] = A[j + 1];
                }
                size--;
                i--; // Adjust index after deletion
            }
            break;
        }
    }
}

};

int main() {
    List list;

    for (int i = 1; i <= 10; i++) {
        list.insertAtEnd(i * 10);
    }
    list.insertAtStart(5);
    list.insertAtStart(6);
    list.insertAtStart(7);
    list.insertBefore(40, 0);
    list.insertBefore(60, 0);
    list.insertBefore(90, 0);
    list.deleteAtStart();
    list.deleteAtEnd();
}

```

```

    cout << "Before 5: ";
    list.print();

    list.deleteBeforeAndAfter(70);

    cout << "After 5: ";
    list.print();

    return 0;
}

```

